

Because **water vapor (\$H_2O\$)** and **ozone (\$O_3\$)** have variable distributions across different layers of the atmosphere, their predictor sets require a deeper mathematical breakdown than the fixed-gas sets. [1]

Instead of measuring simple variations in temperature, these algorithms use equations to track how changes in gas molecule density interact with temperature and the satellite viewing angle.

1. The Water Vapor (\$H_2O\$) Predictor Set

Water vapor is the most challenging gas to parameterize because it is highly variable and exhibits self-broadening behavior at high concentrations (water molecules colliding with each other) and foreign-broadening behavior (water molecules colliding with nitrogen or oxygen). [2, 3, 4, 5, 6]

To linearize these complex quantum mechanics, SARTA uses an advanced **11-term to 13-term polynomial suite**. Let:

- $\Delta T = T_{\text{actual}} - T_{\text{ref}}$ (The layer temperature deviation)
- $W_r = Q_{\text{actual}} / Q_{\text{ref}}$ (The scaling ratio of actual water column mass to the reference profile)
- θ = The satellite zenith angle, where $\sec(\theta)$ represents the path length.

The typical mathematical terms computed for every layer are:

Term	Predictor Equation	Physical Property Modeled
1	$W_r \cdot \sec(\theta)$	Pure slant-path mass abundance
2	$\sqrt{W_r} \cdot \sec(\theta)$	Linearizes strong, highly saturated absorption lines
3	$W_r \cdot \Delta T \cdot \sec(\theta)$	Temperature dependency of basic water lines
4	$(W_r \cdot \sec(\theta))^2$	Models high-density continuum/self-broadening
5	$\sqrt{W_r} \cdot \Delta T \cdot \sec(\theta)$	Saturated line shape changes under temperature

Term	Predictor Equation	Physical Property Modeled
6	$W_r \cdot (\Delta T)^2 \cdot \sec(\theta)$	Non-linear thermal Doppler line shifts
7	$W_r \cdot \sec^2(\theta)$	High-angle geometric line-wing adjustments
8	$W_r^2 \cdot \Delta T \cdot \sec(\theta)$	Thermal sensitivity of high-concentration layers
9	ΔT	Pure local kinetic temperature shift
10	$(\Delta T)^2$	Standard local line-broadening
11	$W_r \cdot \sec(\theta) \cdot \Delta T^3$	Higher-order asymmetry in cold, wet environments

Why the square root ($\sqrt{W_r}$)? Infrared radiation passing through heavy humidity hits a saturation limit. SARTA uses a square-root term to accurately scale transmission changes when a layer is already nearly opaque.

2. The Ozone (O_3) Predictor Set

Ozone behaves differently than water vapor. It is mostly concentrated in a dense layer in the stratosphere and is highly sensitive to UV and infrared cross-section variations driven by stratospheric temperature shifts. [7]

SARTA calculates ozone optical depths using a **10-term polynomial loop**. Let $O_r = V_{\text{actual}} / V_{\text{ref}}$ represent the ozone column ratio. The predictor suite includes:

Term	Predictor Equation	Physical Property Modeled
1	$O_r \cdot \sec(\theta)$	Total ozone molecules along the slant observation line
2	$\sqrt{O_r} \cdot \sec(\theta)$	Scales saturated ozone bands (like the core 9.6\

Term	Predictor Equation	Physical Property Modeled
		$\mu_{\text{m}} \text{ band})$
3	$O_r \cdot \Delta T \cdot \sec(\theta)$	Basic temperature dependence of ozone vibration bands
4	$\sqrt{O_r} \cdot \Delta T \cdot \sec(\theta)$	Saturated band variations under stratospheric temperature swings
5	$O_r \cdot (\Delta T)^2 \cdot \sec(\theta)$	Higher-order thermal Doppler line variations
6	$O_r \cdot \sec^2(\theta)$	Geometric path modifications at extreme sensor viewing edges
7	$(O_r \cdot \sec(\theta))^2$	Tracks high-density layer interactions
8	$\Delta T \cdot \sec(\theta)$	Temperature influence along the line of sight
9	$(\Delta T)^2 \cdot \sec(\theta)$	Symmetrical path-thermal expansion corrections
10	$O_r \cdot \Delta T^3 \cdot \sec(\theta)$	Sharp stratospheric temperature gradient adjustments

The Final Multi-Gas Assembly

SARTA runs these matrix loops simultaneously for every layer. It multiplies the water vapor array by the water vapor coefficients, and the ozone array by the ozone coefficients.

It then sums the results into a single layer value:

$$\tau_{\text{Total}} = \tau_{\text{Fixed}} + \tau_{\text{Water}} + \tau_{\text{Ozone}} + \tau_{\text{Trace}}$$

This final sum is what gets pushed into the vertical integration equation to calculate top-of-atmosphere radiances.

Now that you have seen the exact polynomial layouts for both water vapor and ozone, would you like to see how these equations change when moving from the thermal infrared spectrum down to **shortwave infrared solar reflection spectrum**?

- [1] <https://www.atmo.arizona.edu>
- [2] <https://www.gfdl.noaa.gov>
- [3] <https://www.sciencedirect.com>
- [4] <https://scienceofdoom.com>
- [5] <https://www.nature.com>
- [6] <https://www.preprints.org>
- [7] <https://agupubs.onlinelibrary.wiley.com>